

AWS API Gateway Custom Domain Part 2: Routing Modes

To Begin with the Lab

Summary of the Lab

In this part of the **AWS API Gateway Custom Domain** lesson, the focus was on **routing modes**, which decide how a custom domain chooses the correct API. The video explained the three routing modes available when creating a custom domain: **API mappings only**, **routing rules only**, and **routing rules then API mappings**. It showed how **API mappings only** uses the request path to directly map traffic to an API stage, while **routing rules only** adds conditions such as HTTP headers for more advanced routing. The lab also explained how rule **priority** works when multiple routing rules match the same request, and why **routing rules then API mappings** is useful when you want to keep existing mappings and gradually add condition-based routing. This part completed the custom domain concept by showing how one domain can route traffic to the right API stage based on path and request conditions.

Understanding Routing Modes in API Gateway Custom Domains

Routing mode is the decision layer that determines which API should handle a request that arrives on a custom domain. It exists so one domain like `api.company.com` can serve multiple APIs, stages, or request patterns without confusion. **API mappings only** uses the URL path only, **routing rules only** can evaluate path and HTTP headers, and **routing rules then API mappings** checks rules first and falls back to mappings if no rule matches. This gives you flexibility to keep simple path-based routing or move into more advanced conditional routing without breaking existing setups.

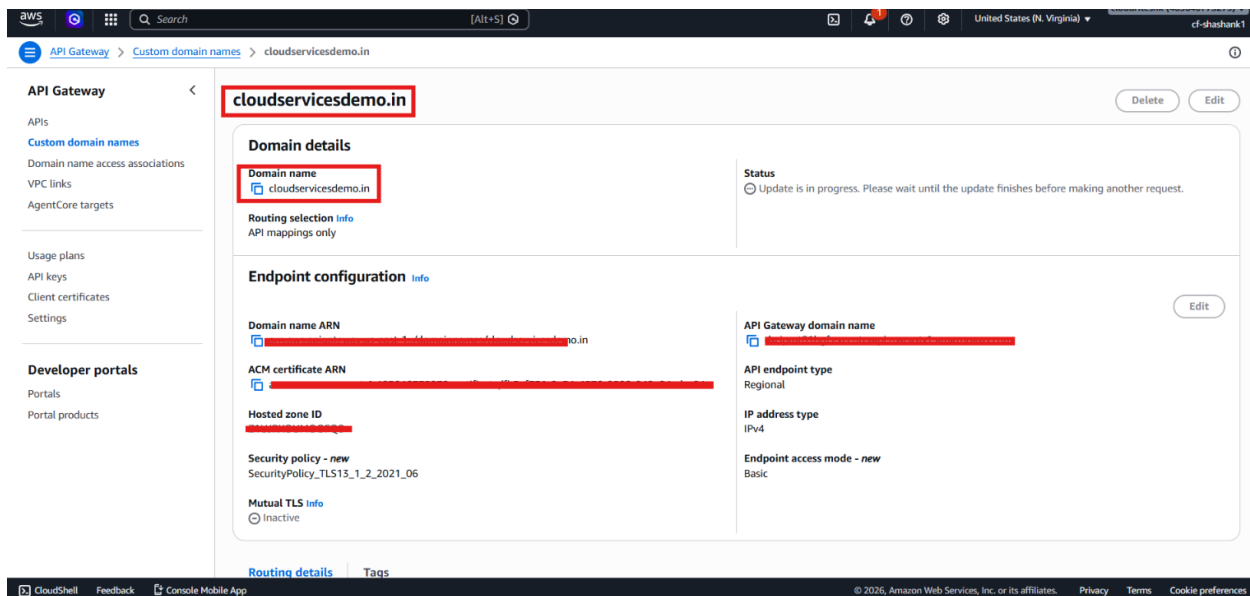
Example:

A company uses `api.company.com/user` to route traffic to UserRestAPI in the dev stage and `api.company.com/prod` to route the same API name to the prod stage. Later, the company wants `/users` to go to dev only when the request contains the header `x-env: dev`, and to prod when the header is `x-env: prod`. Routing rules make that possible, while API mappings alone cannot evaluate headers.

Lab Steps

Step 1 – Review the Existing Custom Domain

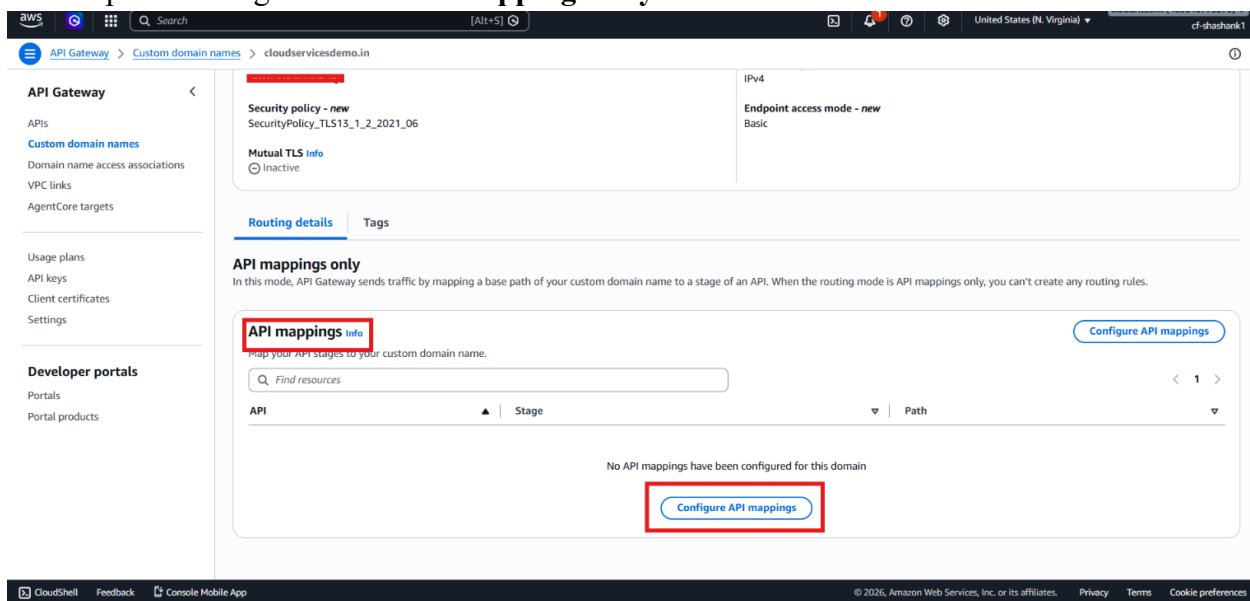
- Sign in to the AWS Management Console.
- Open **API Gateway**.
- Open the existing custom domain created in part one.



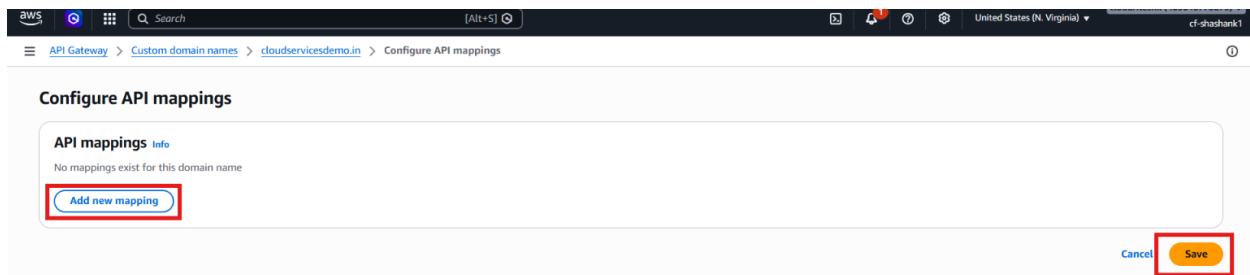
- Review the current routing mode setting.

Step 2 – Use API Mappings Only

- Keep the routing mode as **API mappings only**.



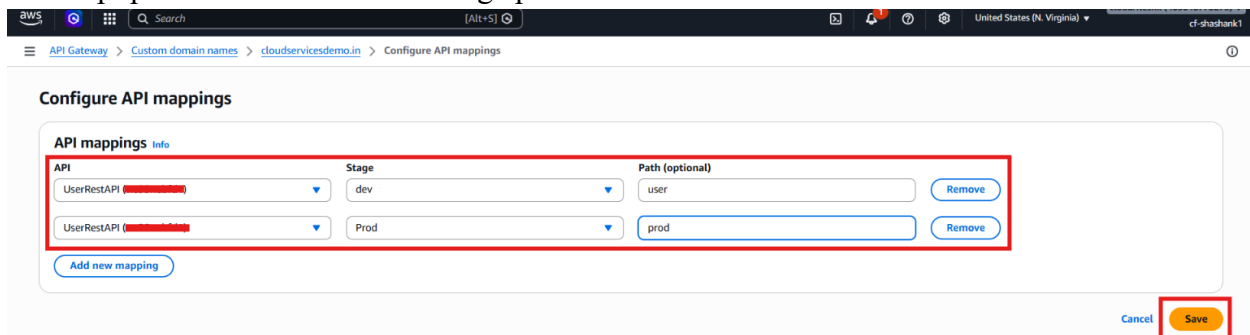
- Add a new mapping rule.



- Select the REST API and stage.
- I have also created a Prod stage so create one if you have not made one yet but only for lab demonstration.
- Confirm that requests are routed only by path.

Step 3 – Understand the Path-Based Example

- Use a custom domain like api.company.com.
- Map /user to UserRestAPI stage dev.
- Map /prod to the same API in stage prod.



- Confirm that the request path determines which API stage is used.

Step 4 – Switch to Routing Rules Only

- Edit the custom domain.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

API Gateway > Custom domain names > cloudservicesdemo.in

API Gateway

- APIs
- Custom domain names
- Domain name access associations
- VPC links
- AgentCore targets

Usage plans

API keys

Client certificates

Settings

Developer portals

Portals

Portal products

Successfully updated API mappings

cloudservicesdemo.in

Delete Edit

Domain details

Domain name
cloudservicesdemo.in

Status
Available

Routing selection Info
API mappings only

Endpoint configuration Info

Domain name ARN
[redacted]demo.in

API Gateway domain name
[redacted]

ACM certificate ARN
[redacted]

API endpoint type
Regional

Hosted zone ID
Z1URXOUM0OFQ8

IP address type
IPv4

Security policy - new
SecurityPolicy_TLS13_1_2_2021_06

Endpoint access mode - new
Basic

Mutual TLS Info
Inactive

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- Change the routing mode to **routing rules only**.
- Save the change.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

API Gateway > Custom domain names > cloudservicesdemo.in > Edit domain details

Edit domain details

Domain details

Domain name
Custom domain names are simpler and more intuitive URLs that you can provide to your API users.

cloudservicesdemo.in

Routing mode - new Info

Select the mode that API Gateway uses to send traffic to your custom domain name. The default mode is API mappings. Changing your routing mode doesn't modify existing routing rules or API mappings.

☐ API mappings only
Connect API stages to your custom domain name.
Supports REST, HTTP, WebSocket APIs.

☒ **Routing rules only (recommended)**
Use routing rules to connect and use path conditions to connect API stages to your custom domain name.
Supports only REST APIs.

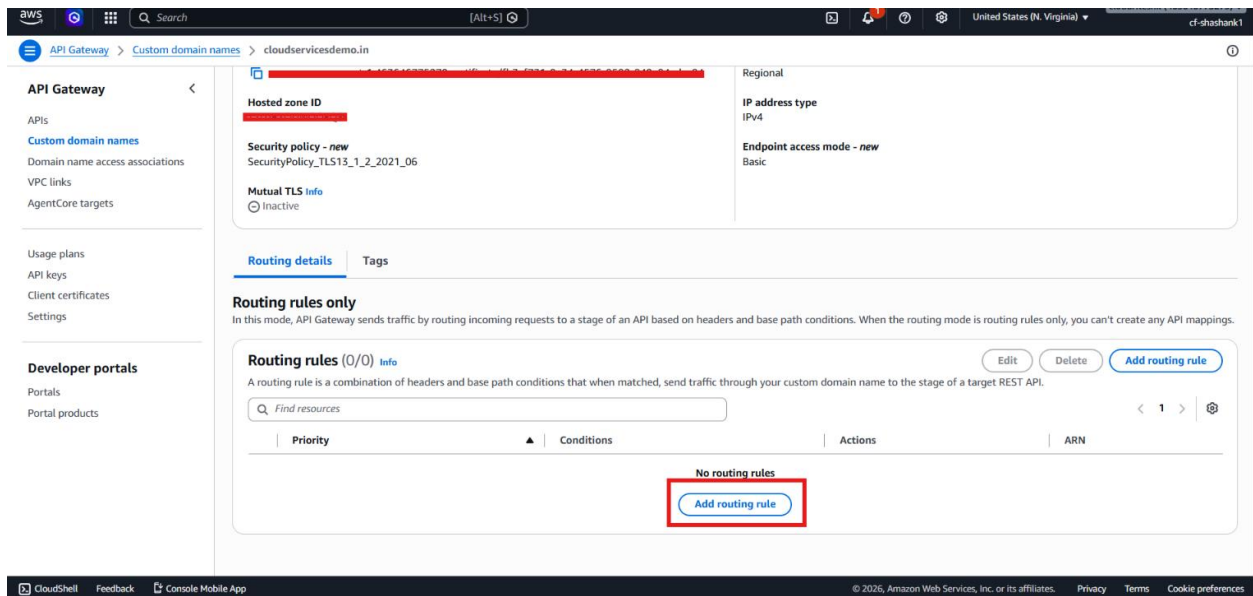
☐ Routing rules then API mappings
Use routing rules and API mappings to connect API stages to your custom domain name. In this mode all routing rules take priority over any API mappings.
Supports only REST APIs.

When your custom domain name uses this mode, you can't view or create any API mappings, but API Gateway retains all API mappings that you created. API callers can't use API mappings to access your custom domain name.

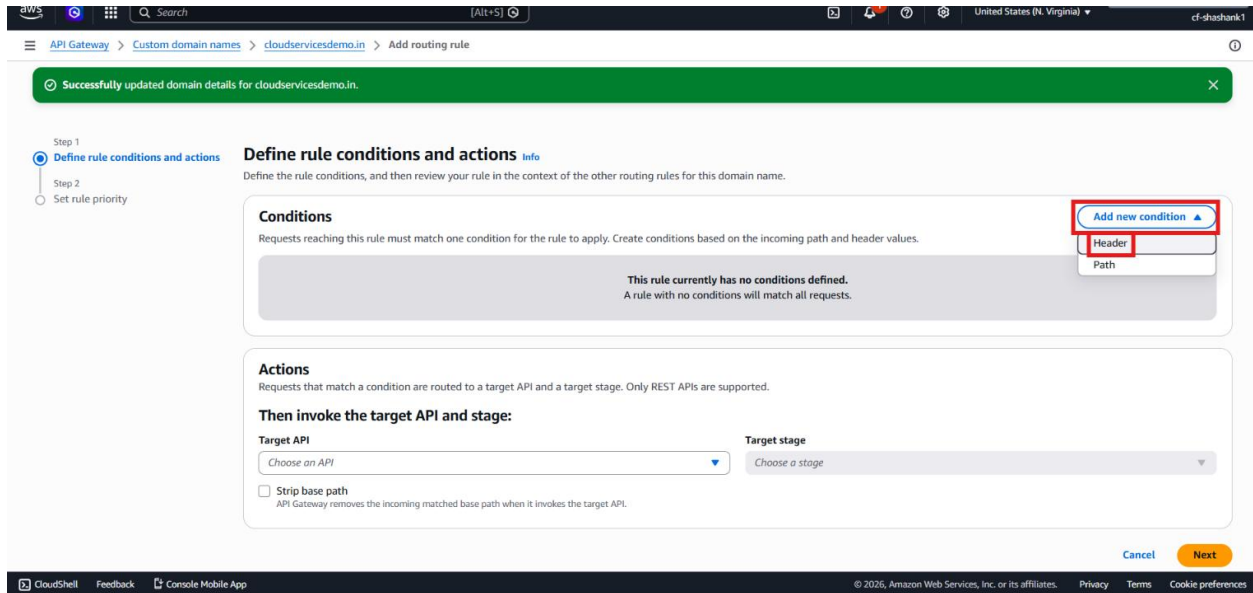
Cancel Save changes

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- Add a new routing rule.



- Choose Path condition.



- Set the user path as shown.

- Choose Header condition and set values as
 - Header – x-env
 - Value – dev
 - Target API – UserRestAPI
 - Target Stage – dev

Step 5 – Set Rule Priority

- Assign a lower number to the more specific rule.
- For this rule I will add 10 for now and create the routing role.

Step 1
Define rule conditions and actions

Step 2
Set rule priority

Set rule priority [Info](#)

Each rule has its own priority and multiple rules can't have the same priority. We recommend that you add gaps between priorities to switch adjacent rules.

Routing rules
Route incoming requests based on incoming paths and headers. Rules are evaluated in priority order from the lowest value to the highest value.

Priority	Conditions	Actions
Priority 10	If request contains header: x-env is dev AND Request matches path: user	Then route to: API: UserRestAPI Stage: dev Strip base path: Inactive

[Cancel](#)
[Previous](#)
[Create routing rule](#)

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API Gateway Custom domain names cloudservicesdemo.in Add routing rule

Step 2
Define rule conditions and actions [Info](#)

Define the rule conditions, and then review your rule in the context of the other routing rules for this domain name.

Conditions
Requests reaching this rule must match one condition for the rule to apply. Create conditions based on the incoming path and header values.

If the request matches this header:

x-env

prod

Remove

AND

If the request matches this path:

cloudservicesdemo.in/ user

Remove

Actions
Requests that match a condition are routed to a target API and a target stage. Only REST APIs are supported.

Then invoke the target API and stage:

Target API

UserRestAPI

Target stage

Prod

☐ Strip base path
API Gateway removes the incoming matched base path when it invokes the target API.

[Cancel](#)
[Next](#)

- I will do the same for another condition as shown here.

- Now I will set its priority to 20.

API Gateway > Custom domain names > cloudservicesdemo.in > Add routing rule

Successfully created routing rule.

Step 1 Define rule conditions and actions
Step 2 **Set rule priority**

Set rule priority [info](#)

Each rule has its own priority and multiple rules can't have the same priority. We recommend that you add gaps between priorities to switch adjacent rules.

Routing rules

Route incoming requests based on incoming paths and headers. Rules are evaluated in priority order from the lowest value to the highest value.

Priority	Conditions	Actions
20	If request contains header: x-env is prod AND Request matches path: user	Then route to: API: UserRestAPI Stage: Prod Strip base path: Inactive
10	If request contains header: x-env is dev AND Request matches path: user	Then route to: API: UserRestAPI Stage: dev Strip base path: Inactive

Cancel Previous **Create routing rule**

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- Note – Lower number means high priority.

Step 6 – Test Rule Matching Behavior

- Review a request that includes the matching path and header.
- Confirm that both a specific and a general rule may match.
- Use priority to ensure the correct rule is chosen.
- Review a request without the header.
- Confirm that the general rule still matches even without a header condition.

Step 7 – Use Routing Rules Then API Mappings

- Edit the routing mode again.
- Select **routing rules then API mappings**.
- Save the change.

API Gateway > Custom domain names > cloudservicesdemo.in > Edit domain details

Edit domain details

Domain details

Domain name
Custom domain names are simpler and more intuitive URLs that you can provide to your API users.

cloudservicesdemo.in

Routing mode - new [info](#)
Select the mode that API Gateway uses to send traffic to your custom domain name. The default mode is API mappings. Changing your routing mode doesn't modify existing routing rules or API mappings.

☐ **API mappings only**
Connect API stages to your custom domain name.
Supports REST, HTTP, WebSocket APIs.

☐ **Routing rules only (recommended)**
Create rules on headers and base path conditions to connect API stages to your custom domain name.
Supports only REST APIs.

☒ **Routing rules then API mappings**
Use routing rules and API mappings to connect API stages to your custom domain name. In this mode all routing rules take priority over any API mappings.
Supports only REST APIs.

Cancel **Save changes**

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- Confirm that routing rules are checked first.
- Confirm that API mappings are used if no routing rule matches.
- Use this mode when you want to add conditional routing without breaking old mappings.

Verification

- The three custom domain routing modes were explained successfully.
- **API mappings only** was understood as path-based routing.
- **Routing rules only** was understood as condition-based routing.
- Rule **priority** was used to resolve matching conflicts.
- **Routing rules then API mappings** was identified as the mixed compatibility mode.
- The custom domain routing logic in **API Gateway** was fully clarified.